DM Learning Objectives Spring 2026

Lecture BCHM	**Glycolysis and PPP and their role in Metabolism**
SOM. MK.I. BPM2.3. DM.2.BCHM.1089	Review the roles of glycolysis and pentose phosphate pathway in metabolism
SOM. MK.I. BPM2.3. DM.2.BCHM.1090	Compare and contrast the action of glucokinase and hexokinase (with the help of a graph)
SOM. MK.I. BPM2.3. DM.2.BCHM.1091	Describe the regulation of glycolysis indicating the regulatory enzymes.
SOM. MK.I. BPM2.3. DM.2.BCHM.1092	Appraise the role of AMP, ATP and fructose 2,6 bisphosphate on glycolysis.
SOM. MK.I. BPM2.3. DM.2.BCHM.1093	Highlight the regulation of pyruvate kinase.
SOM. MK.I. BPM2.3. DM.2.BCHM.1094	Explain the role of glucokinase in regulating blood glucose levels. Predict the effect of mutation in pancreatic glucokinase (MODY-2) on blood glucose levels.
SOM. MK.I. BPM2.3. DM.2.BCHM.1095	Outline the function of glycolysis in specific tissues including liver, brain, muscle, eye, tumor cells.
SOM. MK.I. BPM2.3. DM.2.BCHM.1096	Analyze the effect of arsenic on glycolytic enzymes
SOM. MK.I. BPM2.3. DM.2.BCHM.1097	Identify the biochemical mechanisms involved in the occurrence of lactic acidosis in states of poor tissue perfusion/ oxygenation and metabolic states where aerobic metabolism is hindered.
SOM. MK.I. BPM2.3. DM.2.BCHM.1098	Explain the role of NADPH generated by the pentose phosphate pathway. Highlight its use in fatty acid and cholesterol synthesis
DLA BCHM	**Glycolysis (Reading assignment)**
SOM. MK.I. BPM2.3. DM.2.BCHM.1084	Define glycolysis and explain its role in the generation of metabolic energy.
SOM. MK.I. BPM2.3. DM.2.BCHM.1085	List the reactions and enzymes that convert glucose into pyruvate (emphasize key enzymes and irreversible reactions)
SOM. MK.I. BPM2.3. DM.2.BCHM.1086	Describe substrate level phosphorylation and identify those reactions in glycolytic pathway
SOM. MK.I. BPM2.3. DM.2.BCHM.1087	Explain significance of lactate production in anaerobic glycolysis. Explain the further fate of lactate formed in RBC and muscle and highlight importance of the Cori cycle.
SOM. MK.I. BPM2.3. DM.2.BCHM.1088	Outline the regulation of glycolysis indicating the regulatory enzymes. Highlight the role of AMP, ATP on glycolysis.
DLA BCHM	**Overview of metabolism in the fed fasting state**
SOM. MK.III. BPM2.3. DM.2.BCHM. 1106	Outline the metabolic changes that occur during the feed/fast cycle.
SOM. MK.III. BPM2.3. DM.2.BCHM. 1107	Discuss pathways that are active/inactive in each major organ/tissue during the feed/fast cycle and their regulation and coordination.
DLA BCHM	**Structure and function - Carbohydrates**
SOM. MK.I. BPM2.3. DM.2.BCHM.1099	Outline classification of monosaccharides based on number of carbon atoms and functional group (aldoses and ketoses)
SOM. MK.I. BPM2.3. DM.2.BCHM.1100	Differentiate between: Reducing and non-reducing sugars
SOM. MK.I. BPM2.3. DM.2.BCHM.1101	Define epimer using galactose as an example
SOM. MK.I. BPM2.3. DM.2.BCHM.1102	Compare and contrast the bonds present in: Monosaccharides (glucose, galactose, fructose); Disaccharides (maltose, sucrose, lactose); Polysaccharides (starch, glycogen and cellulose)
SOM. MK.I. BPM2.3. DM.2.BCHM.1103	Identify significance of sugar alcohols (sorbitol, galactitol) and sugar acids (glucuronic acid)
SOM. MK.I. BPM2.3. DM.2.BCHM.1104	Compare sucrose with High Fructose Corn Syrup (HFCS)
SOM. MK.I. BPM2.3. DM.2.BCHM.1105	Review clinical significance of detecting reducing sugars in urine in diabetes mellitus and inherited disorders of fructose and galactose metabolism

Lecture ANAT	**Clinical Embryology of the Gastrointestinal System**
SOM.MK.1.BPM2.3.DM.1.ANAT.1099	Describe the development of the gut tube and the derivatives of the embryonic foregut, midgut and hindgut.
SOM.MK.1.BPM2.3.DM.1.ANAT.1100	Describe the rotation of the foregut, midgut and hindgut during development.
SOM.MK.1.BPM2.3.DM.1.ANAT.1101	Describe the development and clinical picture of duodenal atresia.
SOM.MK.1.BPM2.3.DM.1.ANAT.1102	Compare and contrast the development and clinical picture of congenital hypertrophic pyloric stenosis versus duodenal stenosis
SOM.MK.1.BPM2.3.DM.1.ANAT.1103	Describe the development of the liver, biliary apparatus and pancreas.
SOM.MK.1.BPM2.3.DM.1.ANAT.1104	Describe the embryologic mechanism responsible for formation of annular pancreas and its clinical significance
SOM.MK.1.BPM2.3.DM.1.ANAT.1105	Describe the embryonic process that fails leading to malrotation of the gut
SOM.MK.1.BPM2.3.DM.1.ANAT.1106	Describe the importance of the ligamentum venosum and its significance in fetal circulation.
SOM.MK.1.BPM2.3.DM.1.ANAT.1107	Describe the fate of the omphaloenteric (vitelline) duct.
SOM.MK.1.BPM2.3.DM.1.ANAT.1108	Describe the mechanism for development of colonic aganglionosis (Hirshsprung's disease).
SOM.MK.1.BPM2.3.DM.1.ANAT.1109	Describe the developmental mechanism of imperforate anus, anal agenesis, anal stenosis, membranous atresia of the anus, rectal agenesis, and anorectal agenesis
SOM.MK.1.BPM2.3.DM.1.ANAT.1121	Define the terms dorsal and ventral mesentery
Lecture ANAT	**Anatomy of the Stomach, Liver and Gallbladder**
SOM.MK.1.BPM2.3.DM.1.ANAT.1125	Describe the following ligaments, their embryonic origin, peritoneal coverings and their function: falciform, round ligament of the liver (ligamentum teres hepatis), coronary, triangular, median, medial and lateral umbilical, greater omentum, gastrocolic, gastrosplenic, splenorenal ligaments, lesser omentum (hepatoduodenal and hepatogastric ligaments) and suspensory ligament of the
SOM.MK.1.BPM2.3.DM.1.ANAT.1126	Describe the abdominal esophagus, its blood supply and lymphatic drainage.
SOM.MK.1.BPM2.3.DM.1.ANAT.1127	Describe the nerve pathways for pain in a patient with gastroesophageal reflux disease.
SOM.MK.1.BPM2.3.DM.1.ANAT.1129	Describe sliding and paraesophageal hiatal hernia
SOM.MK.1.BPM2.3.DM.1.ANAT.1130	Describe the anatomy of the stomach, and identify its parts and curvatures.
SOM.MK.1.BPM2.3.DM.1.ANAT.1131	Describe the relationships of the stomach to the spleen, pancreas, liver, ascending and descending colon.
SOM.MK.1.BPM2.3.DM.1.ANAT.1132	Describe the blood supply and the venous and lymphatic drainage of the stomach.
SOM.MK.1.BPM2.3.DM.1.ANAT.1159	Describe the location of the liver and its relationship to surrounding structures.
SOM.MK.1.BPM2.3.DM.1.ANAT.1160	Describe the four anatomical lobes of the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1161	Describe the functional (physiological) lobes of the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1162	List the structures entering / leaving the liver at the porta hepatis.
SOM.MK.1.BPM2.3.DM.1.ANAT.1163	Describe the contents of the hepatoduodenal ligament which will be gripped during a Pringle maneuver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1164	Describe the "the bare area" of the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1165	Describe the blood supply, venous and lymphatic drainage to the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1166	Describe the contents of a liver segment and its importance in surgery of the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1167	Describe the location of the gall bladder and its relationship to the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1168	Describe the anatomy of the biliary tree.
SOM.MK.1.BPM2.3.DM.1.ANAT.1169	Describe the arterial supply, venous and lymphatic drainage of the gall bladder and cystic duct.
SOM.MK.1.BPM2.3.DM.1.ANAT.1172	List the borders of the triangle of Calot and state the importance of this triangle when performing laparoscopic surgery for gallbladder disease.

DLA ANAT	**Autonomics of abdomen**
SOM.MK.1.BPM2.3.DM.1.ANAT.1110	Describe the location of the abdominal sympathetic chain.
SOM.MK.1.BPM2.3.DM.1.ANAT.1111	Describe the location of the sympathetic ganglia associated with the abdominal organs.
SOM.MK.1.BPM2.3.DM.1.ANAT.1112	Describe the contributions from the thoracic splanchnic and lumbar splanchnic nerves to the para- and preaortic sympathetic ganglia.
SOM.MK.1.BPM2.3.DM.1.ANAT.1113	Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.
SOM.MK.1.BPM2.3.DM.1.ANAT.1114	Describe the effect of sympathetic and parasympathetic innervation on the gastrointestinal system.
SOM.MK.1.BPM2.3.DM.1.ANAT.1115	Describe how pain from the abdominal organs may be referred to the body wall.
SOM.MK.1.BPM2.3.DM.1.ANAT.1116	Describe the location of the referred pain from the gastrointestinal tract and organs based on innervation and embryological origin.
DLA ANAT	**Gastrointestinal system and peritonium relationships**
SOM.MK.1.BPM2.3.DM.1.ANAT.1117	Describe the parietal and visceral peritoneum.
SOM.MK.1.BPM2.3.DM.1.ANAT.1118	Describe the extent of the peritoneal cavity, the greater and lesser sacs, and the omental (epiploic) foramen (of Winslow).
SOM.MK.1.BPM2.3.DM.1.ANAT.1119	Describe the difference between intraperitoneal and retroperitoneal abdominal organs.
SOM.MK.1.BPM2.3.DM.1.ANAT.1120	Describe the differences between primary and secondary retroperitoneal abdominal organs.
SOM.MK.1.BPM2.3.DM.1.ANAT.1122	Describe the following peritoneal mesenteries and the abdominal organs they enclose/suspend: “the mesentery”, transverse mesocolon and sigmoid mesocolon.
SOM.MK.1.BPM2.3.DM.1.ANAT.1123	Describe the contents of the supra and infracolic compartments, the location of the paracolic gutters, the vesicouterine pouch, the rectouterine pouch (of Douglas) and the hepatorenal pouch (of Morrison) within the peritoneal cavity.
Lecture BCHM	**Gluconeogenesis**
SOM. MK.I.BPM2.2.DM.3.BCHM.1132	Describe the purpose of gluconeogenesis and indicate the organs and cellular compartments where gluconeogenesis occurs.
SOM. MK.I.BPM2.2.DM.3.BCHM.1133	Outline hepatic gluconeogenesis (irreversible steps, substrates).
SOM. MK.I.BPM2.2.DM.3.BCHM.1134	Justify why there can be no net production of glucose from the carbons of acetyl CoA or from fatty acid degradation leading to acetyl CoA.
SOM. MK.I.BPM2.2.DM.3.BCHM.1135	Discuss the glucose-alanine cycle and the Cori cycle and the formation of pyruvate in the liver.
SOM. MK.I.BPM2.2.DM.3.BCHM.1136	Describe the reaction catalyzed by pyruvate carboxylase (biotin and acetyl CoA as activator).
SOM. MK.I.BPM2.2.DM.3.BCHM.1137	Discuss the reaction catalyzed by PEP carboxykinase (GTP) and glucose 6-phosphatase.
SOM. MK.I.BPM2.2.DM.3.BCHM.1138	Describe how precursors (glucogenic amino acids, lactate and glycerol) are channeled into gluconeogenesis during starvation.
SOM. MK.I.BPM2.2.DM.3.BCHM.1139	Describe the reaction catalyzed by fructose 1,6-bisphosphatase and its inhibition by AMP and F 2,6-bisP.
SOM. MK.I.BPM2.2.DM.3.BCHM.1140	Describe the hepatic bifunctional enzyme and distinguish the role of glucagon and insulin on gluconeogenesis.
SOM. MK.I.BPM2.2.DM.3.BCHM.1141	Explain the effect of glucose 6-phosphatase deficiency in both gluconeogenesis and glycogen degradation.
SOM. MK.I.BPM2.2.DM.3.BCHM.1142	Discuss the importance of beta-oxidation for provision of ATP (energy) and allosteric regulators for gluconeogenesis
SOM. MK.I.BPM2.2.DM.3.BCHM.1143	Outline the role of fatty acid oxidation in providing ATP for gluconeogenesis in liver and highlight the role of acetyl CoA as an allosteric regulator of gluconeogenesis.

Lecture ANAT	**Anatomy of the Pancreas, Spleen and Small Intestine**
SOM.MK.1.BPM2.3.DM.1.ANAT.1125	Describe the following ligaments, their embryonic origin, peritoneal coverings and their function: falciform, round ligament of the liver (ligamentum teres hepatis), coronary, triangular, median, medial and lateral umbilical, greater omentum, gastrocolic, gastrosplenic, splenorenal ligaments, lesser omentum (hepatoduodenal and hepatogastric ligaments) and suspensory ligament of the
SOM.MK.1.BPM2.3.DM.1.ANAT.1173	Describe the location of the pancreas and its relationship to the surrounding structures.
SOM.MK.1.BPM2.3.DM.1.ANAT.1174	Describe the blood supply, venous and lymphatic drainage of the pancreas.
SOM.MK.1.BPM2.3.DM.1.ANAT.1175	Describe how pancreatic enzymes flow to the duodenum.
SOM.MK.1.BPM2.3.DM.1.ANAT.1176	Describe the location of the spleen and its relationship to surrounding structures.
SOM.MK.1.BPM2.3.DM.1.ANAT.1177	Describe the blood supply, and the venous and lymphatic drainage of the spleen.
SOM.MK.1.BPM2.3.DM.1.ANAT.1135	Describe the different portions of the small intestine and explain the external and internal differences between these parts.
SOM.MK.1.BPM2.3.DM.1.ANAT.1136	Describe the location of the duodenum and its relationship to surrounding organs.
SOM.MK.1.BPM2.3.DM.1.ANAT.1137	Describe the location of the small intestines and its relationship to other organs
SOM.MK.1.BPM2.3.DM.1.ANAT.1138	Describe the blood supply and the venous and lymphatic drainage of the duodenum, jejunum and ileum.
SOM.MK.1.BPM2.3.DM.1.ANAT.1147	Descibe intussusception and its clinical presentation
SOM.MK.1.BPM2.3.DM.1.ANAT.1157	Differentiate between small and large bowel obstruction and their clinical presentation
Lecture BCHM	**PDH and TCA cycle**
SOM. MK.I.BPM2.2.DM.3.BCHM.1108	Indicate how pyruvate is transported from the cytosol into the mitochondrial matrix
SOM. MK.I.BPM2.2.DM.3.BCHM.1109	Summarize overall reaction catalyzed by PDH complex and the cofactors that it requires
SOM. MK.I.BPM2.2.DM.3.BCHM.1110	Outline regulation of PDH complex highlighting the role of NADH, acetyl-CoA, phosphorylation of PDH complex.
SOM. MK.I.BPM2.2.DM.3.BCHM.1111	Indicate inhibitory effects of arsenic on PDH
SOM. MK.I.BPM2.2.DM.3.BCHM.1112	Discuss consequences of thiamine deficiency and its effect on PDH and αKGDH complexes in Wernicke-Korsakoff syndrome and beri-beri.
SOM. MK.I.BPM2.2.DM.3.BCHM.1113	Summarize inherited disorders affecting the PDH complex and their metabolic and clinical consequences.
SOM. MK.I.BPM2.2.DM.3.BCHM.1114	Explain the central role of TCA cycle in metabolism, including its catabolic & anabolic functions (amphibolic role)
SOM. MK.I.BPM2.2.DM.3.BCHM.1115	Outline intermediates and enzymes of the TCA cycle
SOM. MK.I.BPM2.2.DM.3.BCHM.1116	Outline energetics of the cycle, including reactions where reducing equivalents (NADH and FADH2) are produced, and the production of GTP by substrate level phosphorylation
SOM. MK.I.BPM2.2.DM.3.BCHM.1117	Explain why aerobic metabolism of glucose using TCA cycle is much more efficient than anaerobic glycolysis
SOM. MK.I.BPM2.2.DM.3.BCHM.1118	Explain regulation of the TCA cycle highlighting the role of NADH, ATP, succinyl-CoA, Ca^{2+} and oxaloacetate.
SOM. MK.I.BPM2.2.DM.3.BCHM.1119	Indicate the effect of fluorocitrate and malonate on TCA cycle
Lecture ANAT	**Anatomy of the Large Intestine**
SOM.MK.1.BPM2.3.DM.1.ANAT.1140	Explain the passage tumor cells may take via the lymphatic drainage from different portions of the small large intestine
SOM.MK.1.BPM2.3.DM.1.ANAT.1142	Explain which parts of the GI tract may become involved in volvulus.
SOM.MK.1.BPM2.3.DM.1.ANAT.1146	Explain the surgical approach through the layers of the abdominal wall to approach an inflamed appendix and the surgical significance of the muscle fiber orientation
SOM.MK.1.BPM2.3.DM.1.ANAT.1148	Describe the location of the large intestine and its relationship to other organs.

SOM.MK.1.BPM2.3.DM.1.ANAT.1149	Describe the different portions of the large intestine and explain the external and internal differences between these parts.
SOM.MK.1.BPM2.3.DM.1.ANAT.1150	Describe the blood supply and the venous and lymphatic drainage of the large intestine.
SOM.MK.1.BPM2.3.DM.1.ANAT.1151	Describe the location, arterial supply, and the venous and lymphatic drainage of the rectum and anal canal.
SOM.MK.1.BPM2.3.DM.1.ANAT.1152	Describe the location of the pectinate line and the changes that occur at the pectinate line including arterial supply, venous and lymphatic drainage, innervation and embryological origin.
SOM.MK.1.BPM2.3.DM.1.ANAT.1153	Describe the anastomotic communications between vessels of the foregut and the midgut and between midgut and hindgut.
SOM.MK.1.BPM2.3.DM.1.ANAT.1155	Explain why surgical resection of the ascending or descending colon should be done in the lateral paracolic gutters.
SOM.MK.1.BPM2.3.DM.1.ANAT.1156	Describe the structural modifications and clinical signs of diverticulosis.
SOM.MK.1.BPM2.3.DM.1.ANAT.1157	Differentiate between small and large bowel obstruction and their clinical presentation
SOM.MK.1.BPM2.3.DM.1.ANAT.1158	Define enterocele and rectocele
SOM.MK.1.BPM2.3.DM.1.ANAT.1181	Describe the portal circulation and the areas of anastomosis with the systemic circulation
SOM.MK.1.BPM2.3.DM.1.ANAT.1154	Explain how the hind gut receives its blood supply after repair of an abdominal aortic aneurysm since the inferior mesenteric artery is usually sacrificed during the procedure.
Lecture BCHM	**Electron Transport Chain and Oxidative phosphorylation**
SOM. MK.I.BPM2.2.DM.3.BCHM.1120	Identify clinical features of mitochondrial disorders
SOM. MK.I.BPM2.2.DM.3.BCHM.1121	List diseases involving mutations in mitochondrial DNA: Leber Hereditary Optic Neuropathy, Myoclonic Epilepsy and Ragged-Red Fiber Disease, MELAS
SOM. MK.I.BPM2.2.DM.3.BCHM.1122	Outline the aspartate-malate and glycerophosphate (primarily white muscle tissue) shuttles and their significance in regenerating cytoplasmic NAD+
SOM. MK.I.BPM2.2.DM.3.BCHM.1123	Outline the different types of electron carriers found in the ETC complexes and follow the path electrons from NADH and FADH2 to oxygen
SOM. MK.I.BPM2.2.DM.3.BCHM.1124	Discuss the mode of action/effects of specific inhibitors of the ETC (including antimycin A, azide, hydrogen sulphide, cyanide, rotenone, piericidin A, carbon monoxide)
SOM. MK.I.BPM2.2.DM.3.BCHM.1125	Differentiate between: direct inhibitors of the ETC, uncouplers of oxidative phosphorylation, ATP synthase inhibitors (oligomycin) and inhibitors of the ATP/ADP translocase
SOM. MK.I.BPM2.2.DM.3.BCHM.1126	Explain how the electrochemical gradient is generated during electron transport to drive ATP synthesis and consider Mitchell's Chemiosmotic theory
SOM. MK.I.BPM2.2.DM.3.BCHM.1127	Describe the mode of action of mitochondrial ATP synthase and its inhibition by oligomycin
SOM. MK.I.BPM2.2.DM.3.BCHM.1128	Define uncoupling of oxidative phosphorylation, natural uncoupling in brown adipose tissue versus other types of uncouplers (2,4-DNP)
SOM. MK.I.BPM2.2.DM.3.BCHM.1129	Discuss ADP-ATP translocase and inhibitors of the transporter (atractyloside and bongkrekic acid)
SOM. MK.I.BPM2.2.DM.3.BCHM.1130	Discuss respiratory control of the ETC and the P/O ratio
SOM. MK.I.BPM2.2.DM.3.BCHM.1131	Identify the role of oxidative phosphorylation at conditions of low oxygen
Lecture ANAT	**Imaging of the Gastrointestinal System**
SOM.MK.1.BPM2.3.DM.1.ANAT.1141	Identify the various parts of the GI tract and their blood supply in CT, MRI and radiological images.

SOM.MK.1.BPM2.3.DM.1.ANAT.1178	Identify the liver, pancreas and spleen, their blood supply and other adjacent abdominal structures in CT, MRI and radiological images
Lecture BCHM	**Glycogen Metabolism and Glycogen Storage Diseases**
SOM. MK.I.BPM2.3.DM.2.BCHM.1154	Describe the structure-function relationship of glycogen in tissues that store glycogen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1155	Compare and contrast the purpose of glycogen metabolism in skeletal muscle and liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1156	Describe the role of insulin in regulation of glycogenesis (glycogen synthase) in liver and muscle.
SOM. MK.I.BPM2.3.DM.2.BCHM.1157	Outline glycogenesis and review the significance of glycogen synthase and the branching enzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1158	Compare and contrast glycogen degradation in liver and muscle with reference to: hormones, metabolic state, and fate of glucose 6-P.
SOM. MK.I.BPM2.3.DM.2.BCHM.1159	Outline glycogenolysis and highlight importance of glycogen phosphorylase and debranching enzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1160	Evaluate the role of insulin and glucagon in reciprocal regulation of carbohydrate metabolism in the liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1161	Compare and contrast carbohydrate metabolism in the liver during fed and fasting state.
SOM. MK.I.BPM2.3.DM.2.BCHM.1162	List the glycogen storage diseases (Types I through VII) and be able to differentiate the disorders based on clinical and laboratory findings.
SOM. MK.I.BPM2.3.DM.2.BCHM.1163	Describe type I (von Gierke) glycogen storage disease including the metabolic basis for severe hypoglycemia, hepatomegaly and lactic acidosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1164	Describe type II (Pompe disease) and outline lysosomal glycogen degradation
SOM. MK.I.BPM2.3.DM.2.BCHM.1165	Differentiate between glycogen storage diseases type III (Cori) and IV (Andersen) based on the abnormal structure of glycogen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1166	Explain basis for clinical manifestations of type V (McArdle) glycogen storage disease: decreased exercise tolerance and lack of lactate increase in blood after exercise.
Lecture ANAT	**Flipped Class: Clinical Gastrointestinal System**
SOM.MK.1.BPM2.3.DM.1.ANAT.1124	Explain where free air and fluid in the peritoneal cavity usually accumulates.
SOM.MK.1.BPM2.3.DM.1.ANAT.1128	Describe achalasia and its clinical presentation
SOM.MK.1.BPM2.3.DM.1.ANAT.1133	Compare and contrast where stomach contents from a perforation of the posterior, versus the anterior stomach wall will initially accumulate.
SOM.MK.1.BPM2.3.DM.1.ANAT.1134	List the blood vessels which may be eroded in ulceration of the posterior wall of the stomach.
SOM.MK.1.BPM2.3.DM.1.ANAT.1139	Identify and describe the large artery at risk for injury/hemorrhage by a perforated ulcer in the posterior wall of the duodenal bulb (1st part of duodenum).
SOM.MK.1.BPM2.3.DM.1.ANAT.1143	Describe the anatomical location, mechanism and clinical sequences of renal vein entrapment syndrome (review from CPR)
SOM.MK.1.BPM2.3.DM.1.ANAT.1144	Describe the psoas sign and explain why it can be positive during appendicitis.
SOM.MK.1.BPM2.3.DM.1.ANAT.1145	Explain the surgical significance of Mc Burney's point.
SOM.MK.1.BPM2.3.DM.1.ANAT.1179	Describe the anatomical location, mechanism and clinical sequences of the insertion of a nasogastric tube.
SOM.MK.1.BPM2.3.DM.1.ANAT.1170	List the 3 common sites where gall stones can be impacted (lodged) and explain the effects of each on the bile flow.
SOM.MK.1.BPM2.3.DM.1.ANAT.1171	Explain how the visceral pain from an inflamed gallbladder can refer to the shoulder.
SOM.MK.1.BPM2.3.DM.1.ANAT.1180	Explain the pathway taken by an endoscope from mouth to the biliary tree during Endoscopic Retrograde Cholangio Pancreatography (ERCP) to remove a gallstone

SOM.MK.1.BPM2.3.DM.1.ANAT.1182	Describe the anatomical basis for the clinical symptoms of portal hypertension including esophageal varices, rectal varices and caput medusa
SOM.MK.1.BPM2.3.DM.1.ANAT.1183	Explain how portal hypertension can be relieved by portacaval shunts
Lecture BCHM	**Regulation of Carbohydrate Metabolism**
SOM. MK.I.BPM2.2.DM.2.BCHM.1144	Discuss the concept of reciprocal regulation of glycolysis and gluconeogenesis in the fed state and during fasting.
SOM. MK.I.BPM2.2.DM.2.BCHM.1145	Explain the use of glycogen as a source of glucose in skeletal muscle and outline regulation of glycolysis in muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1146	Analyze the role of calcium and AMP in regulation of glycogenolysis in skeletal muscle
SOM. MK.I.BPM2.2.DM.2.BCHM.1147	Describe the effect of epinephrine on muscle contraction.
SOM. MK.I.BPM2.2.DM.2.BCHM.1148	Describe metabolic changes in the muscle during exercise and describe ATP formation in muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1149	Review regulation of glucokinase and allosteric regulation of hepatic glycolysis (PFK-1 and feed-forward activation of pyruvate kinase).
SOM. MK.I.BPM2.2.DM.2.BCHM.1150	Describe the role of fructose 2,6-bisphosphate in reciprocal regulation of glycolysis and gluconeogenesis in the liver.
SOM. MK.I.BPM2.2.DM.2.BCHM.1152	Assess regulation of glycogen metabolism by covalent modification and allosteric modulators in liver and muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1153	Review regulation of hepatic carbohydrate metabolism after a meal and compare it to the fasting state.
Small group HPWP	**DM SG 01 - BPM2 HPWP 03 -Social Determinants of Health: Examining Systematic Impacts on Vulnerable Populations**
SOM.MK.3.BPM2.HPWP.035	Compare and contrast unique social determinants of health related to pregnancy and nursing healthcare e.g., increased risk caused by inequities
SOM.MK.2.BPM2.HPWP.036	Identify the leading causes of maternal mortality and morbidity in the U.S., and the risk factors that are responsible for them. e.g. race, socioeconomic status
SOM.CS.1.BPM2.HPWP.037	Explain how the patient encounter in pregnancy and nursing healthcare is negatively impacted by implicit biases about patient ethnicity/race, appearance, perceived social status, and language
SOM.PB.6.BPM2.HPWP.038	Create a set of culturally-responsive guidelines for working with Black pregnant and nursing persons.
SOM.MK.3.BPM2.HPWP.039	Compare and contrast unique risk factors for health conditions related to sexual orientation and gender identity e.g., not appropriately capturing relevant sexual history
SOM.CS.1.BPM2.HPWP.040	Identify some of the challenges transgender women/men encounter when presenting for care
SOM.CS.1.BPM2.HPWP.041	Describe actions healthcare professionals can take to optimize patients' comfort levels. e.g. ask the patient their preferred name
SOM.CS.7.BPM2.HPWP.042	Develop systems changes that need to be made in a healthcare setting to create a welcoming care environment for transgender men/women
SOM.PB.3.BPM2.HPWP.043	Evaluate physicians' implicit biases about gender and how they can be managed.
SOM.PB.3.BPM2.HPWP.044	Discuss the physician's role as an advocate and how allyship can improve patient care. e.g., build trust and reduce health disparities
Small group HPWP	**Selecting and Appraising Data for Self-Directed Learning**
SOM.PB.3.BPM2.HPWP.045	Explain self-directed learning (SDL) by defining its characteristics, benefits and challenges.
SOM.PB.3.BPM2.HPWP.046	Apply the PICO framework to create appropriate clinical questions for a chosen scenario.
SOM.PB.3.BPM2.HPWP.047	Develop search strategies by identifying key concepts, selecting appropriate databases, and refining searches to retrieve high-quality information.

SOM.PB.3.BPM2.HPWP.048	Appraise research articles by assessing their reliability, relevance to the question posed, and design quality and justify whether the evidence provided is appropriate for addressing complex clinical questions.
SOM.PB.3.BPM2.HPWP.049	Describe the key tenets of evidence-based medicine (EBM) that inform decision-making.
Lecture HCB	**Histology of the Upper Gastrointestinal Tract**
SOM.MK.I.BPM2.3.DM.1.HCB.1195	Identify the microscopic structure of the lingual papillae.
SOM.MK.I.BPM2.3.DM.1.HCB.1196	Describe the microscopic structure of taste buds.
SOM.MK.I.BPM2.3.DM.1.HCB.1197	Recap the distinct four layers characteristic of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1198	Identify the location of Meissner’s plexus in the wall of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1199	Describe the microscopic structure of Meissner’s plexus in the wall of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1200	Identify the location of Auerbach’s plexus in the wall of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1201	Describe the microscopic structure of Auerbach’s plexus in the wall of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1202	Identify the specific locations of serosa in relation to the GI tract.
SOM.MK.I.BPM2.3.DM.1.HCB.1203	Identify the specific locations of adventitia in relation to the GI tract.
SOM.MK.I.BPM2.3.DM.1.HCB.1204	State the layers of the wall of the esophagus.
SOM.MK.I.BPM2.3.DM.1.HCB.1205	List the components found in the layers of the wall of the esophagus.
SOM.MK.I.BPM2.3.DM.1.HCB.1206	Describe the microscopic structure of the esophagus.
SOM.MK.I.BPM2.3.DM.1.HCB.1207	Differentiate between esophageal cardiac and proper glands based on their location and type of secretions.
SOM.MK.I.BPM2.3.DM.1.HCB.1208	Describe the structure of the muscularis externa throughout the length of the esophagus.
SOM.MK.I.BPM2.3.DM.1.HCB.1209	Describe the structure of the esophagogastric junction.
SOM.MK.I.BPM2.3.DM.1.HCB.1210	Describe the change in the epithelium of the lower esophagus resulting from chronic acid reflux (Barrett’s Esophagus)
SOM.MK.I.BPM2.3.DM.1.HCB.1211	Describe the structure of the three regions of the stomach.
SOM.MK.I.BPM2.3.DM.1.HCB.1212	State the layers of the wall of the stomach
SOM.MK.I.BPM2.3.DM.1.HCB.1213	List the components found in the layers of the wall of the stomach
SOM.MK.I.BPM2.3.DM.1.HCB.1214	Describe the microscopic structure of the gastric mucosa.
SOM.MK.I.BPM2.3.DM.1.HCB.1215	Differentiate between the cells of the gastric mucosa based on their histological structures, functions and localization.
SOM.MK.I.BPM2.3.DM.1.HCB.1216	Differentiate between cardiac, fundic and pyloric glands based on their microscopic structures, locations and type of secretions.
SOM.MK.I.BPM2.3.DM.1.HCB.1217	Identify rugae of the stomach
SOM.MK.I.BPM2.3.DM.1.HCB.1218	State the function(s) of rugae.
SOM.MK.I.BPM2.3.DM.1.HCB.1219	Describe the structure of a rugae.
SOM.MK.I.BPM2.3.DM.1.HCB.1220	State the function of the muscularis externa of the stomach.
SOM.MK.I.BPM2.3.DM.1.HCB.1221	Describe the structure of the muscularis externa of the stomach.
DLA HCB	**Histology of the Gastrointestinal tract Overview**
SOM.MK.I.BPM2.3.DM.1.HCB.1187	Describe the distinct four layers characteristic of the alimentary canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1188	Differentiate between Meissner's and Auerbach's plexuses based on their microscopic structure, function, localization and origins.

SOM.MK.I.BPM2.3.DM.1.HCB.1189	Compare serosa vs. adventitia.
Lecture BCHM	**Metabolism of Fructose and Galactose**
SOM. MK.I.BPM2.3.DM.2.BCHM.1167	Describe the pathway by which dietary fructose enters glycolysis highlighting the role of fructokinase and Aldolase B.
SOM. MK.I.BPM2.3.DM.2.BCHM.1168	Compare and contrast laboratory findings, enzyme deficiency, clinical manifestations and management of Essential Fructosuria and Hereditary Fructose Intolerance.
SOM. MK.I.BPM2.3.DM.2.BCHM.1169	Describe the polyol pathway, its function and its importance in diabetic patients.
SOM. MK.I.BPM2.3.DM.2.BCHM.1170	Describe the metabolism and fate of dietary galactose.
SOM. MK.I.BPM2.3.DM.2.BCHM.1171	Compare and contrast laboratory findings, enzyme deficiency, clinical manifestations and management of Classical and non-classical galactosemia.
SOM. MK.I.BPM2.3.DM.2.BCHM.1172	Differentiate benign fructosuria, galactosemia, lactose intolerance and hereditary fructose intolerance based on clinical features, laboratory findings and dietary management.
Lecture HCB	**Histology of the Lower Gastrointestinal Tract**
SOM.MK.I.BPM2.3.DM.1.HCB.1222	Identify the gastroduodenal junction.
SOM.MK.I.BPM2.3.DM.1.HCB.1223	Describe the microscopic structure of the gastroduodenal junction.
SOM.MK.I.BPM2.3.DM.1.HCB.1224	Describe the changes to the wall of the stomach in the development of ulcers.
SOM.MK.I.BPM2.3.DM.1.HCB.1225	Describe the main complications of chronic peptic ulceration.
SOM.MK.I.BPM2.3.DM.1.HCB.1226	Identify the three anatomical regions of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1227	Give the functions of the three anatomical regions of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1228	Describe the microscopic structure of the three anatomical regions of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1229	Give the function of microvilli, villi and plicae circulares.
SOM.MK.I.BPM2.3.DM.1.HCB.1230	Compare the microscopic structure of microvilli, villi and plicae circulares.
SOM.MK.I.BPM2.3.DM.1.HCB.1231	Identify the mucosa of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1232	Give the function(s) of the mucosa of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1233	Describe the microscopic structure of the small intestinal mucosa.
SOM.MK.I.BPM2.3.DM.1.HCB.1234	List the different cell types of the mucosa of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1235	Distinguish between the cells of the small intestinal mucosa based on location, function and microscopic features
SOM.MK.I.BPM2.3.DM.1.HCB.1236	Give the function(s) of the muscularis externa of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1237	Microscopically describe the structure of the muscularis externa of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1238	Give the function of the Peyer’s patches of the ileum.
SOM.MK.I.BPM2.3.DM.1.HCB.1239	Identify the Peyer’s patches of the ileum.
SOM.MK.I.BPM2.3.DM.1.HCB.1240	Describe the microscopic structure of the Peyer’s patches of the ileum.
SOM.MK.I.BPM2.3.DM.1.HCB.1241	Describe the microscopic differences typical to each of the three parts of the small intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1242	Describe the primary microscopic changes noticed in the GI tract in Celiac disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1243	Describe the primary microscopic changes noticed in the GI tract in Crohn's disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1244	Identify the anatomical regions of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1245	Give the functions of the anatomical regions of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1246	Describe the microscopic structure of the different anatomical regions of the large intestine.

SOM.MK.I.BPM2.3.DM.1.HCB.1247	Identify the mucosa of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1248	Give the function of the mucosa of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1249	Describe the microscopic structure of the large intestinal mucosa.
SOM.MK.I.BPM2.3.DM.1.HCB.1250	List the different cell types of the mucosa of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1251	Distinguish between the cells of the large intestinal mucosa based on location, function and microscopic features
SOM.MK.I.BPM2.3.DM.1.HCB.1252	Describe the microscopic differences typically seen in each of the parts of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1253	Give the function(s) of the muscularis externa of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1254	Microscopically describe the structure of the muscularis externa of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1255	Give the function(s) of the tenia coli.
SOM.MK.I.BPM2.3.DM.1.HCB.1256	Describe the histological structure of the tenia coli .
SOM.MK.I.BPM2.3.DM.1.HCB.1257	Give the function(s) of haustra.
SOM.MK.I.BPM2.3.DM.1.HCB.1258	Describe the histological structure of haustra.
SOM.MK.I.BPM2.3.DM.1.HCB.1259	Describe the histological appearance of an adenomatous polyp of the large intestine.
SOM.MK.I.BPM2.3.DM.1.HCB.1260	Give the function of the appendix.
SOM.MK.I.BPM2.3.DM.1.HCB.1261	Describe the histological appearance of the structure of the appendix.
SOM.MK.I.BPM2.3.DM.1.HCB.1262	Give the function of the rectum.
SOM.MK.I.BPM2.3.DM.1.HCB.1263	Describe the histological appearance of the structure of the rectum.
SOM.MK.I.BPM2.3.DM.1.HCB.1264	Describe the histological appearance of the structure (including the 3 zones) of the anal canal.
SOM.MK.I.BPM2.3.DM.1.HCB.1265	Describe the changes in the autonomic nerve plexus in the wall of the GIT that may lead to Hirschsprung disease (congenital mega colon).
Lecture BCHM	**Fatty Acid Synthesis and Regulation**
SOM. MK.I.BPM2.3.DM.2.BCHM.1181	Indicate the main sites in the body and the metabolic condition under which fatty acid de novo synthesis takes place.
SOM. MK.I.BPM2.3.DM.2.BCHM.1182	Outline the actions of citrate lyase and the malic enzyme in the liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1183	Explain how acetyl-CoA carboxylase (biotin) synthesizes malonyl-CoA.
SOM. MK.I.BPM2.3.DM.2.BCHM.1184	Outline the fatty acid synthetic pathway and indicate the role of the acyl carrier protein in the pathway
SOM. MK.I.BPM2.3.DM.2.BCHM.1185	Outline the denovo synthesis of fatty acids highlighting the significance of acetyl CoA carboxylase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1186	Identify the sources of NADPH for fatty acid synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1187	Outline the elongation and desaturation of fatty acids in humans.
SOM. MK.I.BPM2.3.DM.2.BCHM.1188	Explain the conversion of linoleic acid to arachidonic acid and α-linolenic acid to docosahexaenoic acid.
SOM. MK.I.BPM2.3.DM.2.BCHM.1189	List the major tissues of triacylglycerol synthesis and storage
SOM. MK.I.BPM2.3.DM.2.BCHM.1190	Distinguish between the glycerol-3-P pathway and the MAG pathway for triacylglycerol synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1191	Explain the regulation of acetyl-CoA carboxylase (short-term and long-term) and regulation of Fatty Acid synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1192	Discuss malonyl-CoA inhibition of carnitine-palmitoyl transferase I (CPT I).
Small group ANAT	**DM SG 03 - DM Anatomy**
SOM.MK.II.BPM1.1.DM.1.ANAT.5001	Analyze the anatomy of the stomach and evaluate its parts and curvatures in terms of function and location.
SOM.MK.II.BPM1.1.DM.1.ANAT.5002	Evaluate the spatial relationships and interactions between the stomach, spleen, pancreas, liver, and the ascending and descending colon.

SOM.MK.II.BPM1.1.DM.1.ANAT.5003	Analyze the blood supply, venous drainage, and lymphatic drainage of the stomach, explaining their roles in its function.
SOM.MK.II.BPM1.1.DM.1.ANAT.5004	Assess the development of the gut tube and synthesize the derivatives of the embryonic foregut, midgut, and hindgut.
SOM.MK.II.BPM1.1.DM.1.ANAT.5005	Compare and contrast the accumulation patterns of stomach contents from perforations in the posterior versus the anterior stomach wall.
SOM.MK.II.BPM1.1.DM.1.ANAT.5006	Identify and analyze the blood vessels that could be eroded in cases of ulceration of the posterior stomach wall.
SOM.MK.II.BPM1.1.DM.1.ANAT.5007	Analyze the sympathetic, parasympathetic, and sensory innervation of the foregut, midgut, and hindgut, evaluating their roles in gastrointestinal function.
SOM.MK.II.BPM1.1.DM.1.ANAT.5008	Analyze how pain from abdominal organs is referred to the body wall, and evaluate the mechanisms behind this phenomenon.
SOM.MK.II.BPM1.1.DM.1.ANAT.5009	Examine the location of referred pain from the gastrointestinal tract and organs, relating it to innervation and embryological origin.
SOM.MK.II.BPM1.1.DM.1.ANAT.5010	Explain the surgical significance of McBurney’s point, discussing its relevance in clinical practice.
SOM.MK.II.BPM1.1.DM.1.ANAT.5011	Evaluate the surgical approach through the layers of the abdominal wall for accessing an inflamed appendix, assessing the significance of muscle fiber orientation in the procedure.
SOM.MK.II.BPM1.1.DM.1.ANAT.5012	Analyze the blood supply, venous drainage, and lymphatic drainage of the large intestine, explaining their functional importance.
SOM.MK.II.BPM1.1.DM.1.ANAT.5013	Analyze the anatomy of the biliary tree, assessing its structure and function in relation to bile secretion and transport.
SOM.MK.II.BPM1.1.DM.1.ANAT.5014	Evaluate the location of the pancreas and examine its relationship to surrounding structures, highlighting clinical implications.
SOM.MK.II.BPM1.1.DM.1.ANAT.5015	Explain the process through which pancreatic enzymes flow to the duodenum, identifying key anatomical features involved in the process.
SOM.MK.II.BPM1.1.DM.1.ANAT.5016	Analyze the location of the gallbladder and evaluate its relationship to the liver, emphasizing functional and clinical relevance.
SOM.MK.II.BPM1.1.DM.1.ANAT.5017	Examine the blood supply, venous drainage, and lymphatic drainage of the pancreas, assessing their roles in pancreatic function and health.
SOM.MK.II.BPM2.1.DM.1.ANAT.5001	Analyze the anatomy of the stomach and evaluate its parts and curvatures in terms of function and location.
SOM.MK.II.BPM2.1.DM.1.ANAT.5002	Evaluate the spatial relationships and interactions between the stomach, spleen, pancreas, liver, and the ascending and descending colon.
SOM.MK.II.BPM2.1.DM.1.ANAT.5003	Analyze the blood supply, venous drainage, and lymphatic drainage of the stomach, explaining their roles in its function.
SOM.MK.II.BPM2.1.DM.1.ANAT.5004	Assess the development of the gut tube and synthesize the derivatives of the embryonic foregut, midgut, and hindgut.
SOM.MK.II.BPM2.1.DM.1.ANAT.5005	Compare and contrast the accumulation patterns of stomach contents from perforations in the posterior versus the anterior stomach wall.
SOM.MK.II.BPM2.1.DM.1.ANAT.5006	Identify and analyze the blood vessels that could be eroded in cases of ulceration of the posterior stomach wall.
SOM.MK.II.BPM2.1.DM.1.ANAT.5007	Analyze the sympathetic, parasympathetic, and sensory innervation of the foregut, midgut, and hindgut, evaluating their roles in gastrointestinal function.

SOM.MK.II.BPM2.1.DM.1.ANAT.5008	Analyze how pain from abdominal organs is referred to the body wall, and evaluate the mechanisms behind this phenomenon.
SOM.MK.II.BPM2.1.DM.1.ANAT.5009	Examine the location of referred pain from the gastrointestinal tract and organs, relating it to innervation and embryological origin.
SOM.MK.II.BPM2.1.DM.1.ANAT.5010	Explain the surgical significance of McBurney's point, discussing its relevance in clinical practice.
SOM.MK.II.BPM2.1.DM.1.ANAT.5011	Evaluate the surgical approach through the layers of the abdominal wall for accessing an inflamed appendix, assessing the significance of muscle fiber orientation in the procedure.
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SOM.MK.II.BPM2.1.DM.1.ANAT.5016	Analyze the location of the gallbladder and evaluate its relationship to the liver, emphasizing functional and clinical relevance.
SOM.MK.II.BPM2.1.DM.1.ANAT.5017	Examine the blood supply, venous drainage, and lymphatic drainage of the pancreas, assessing their roles in pancreatic function and health.
Lecture HCB	**Flipped Class: Histology of the Gastrointestinal Glands**
SOM.MK.I.BPM2.3.DM.1.HCB.1266	List the extrinsic glands of the digestive system.
SOM.MK.I.BPM2.3.DM.1.HCB.1267	Give the function(s) of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1268	Give the localization of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1269	Compare exocrine vs. endocrine functions of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1270	Describe the microscopic structure of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1271	Identify the Glisson's capsule of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1272	Describe the flow of blood in the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1273	Differentiate between the classic hepatic lobule, portal lobule and liver acinus based on microscopic structure and function.
SOM.MK.I.BPM2.3.DM.1.HCB.1274	Identify the components of the portal triad.
SOM.MK.I.BPM2.3.DM.1.HCB.1275	Identify the hepatocytes of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1276	Give the function(s) of the hepatocytes of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1277	Give the localization of the hepatocytes of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1278	Describe the microscopic structure of the hepatocytes of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1279	Identify the Ito cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1280	Give the function(s) of the Ito cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1281	Give the localization of the Ito cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1282	Describe the microscopic structure of the Ito cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1283	Identify the Kupffer cells of the liver.

SOM.MK.I.BPM2.3.DM.1.HCB.1284	Give the function(s) of the Kupffer cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1285	Give the localization of the Kupffer cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1286	Describe the microscopic structure of the Kupffer cells of the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1287	Describe the flow of bile in the liver.
SOM.MK.I.BPM2.3.DM.1.HCB.1288	Identify the perisinusoidal space (of Disse).
SOM.MK.I.BPM2.3.DM.1.HCB.1289	Give the function of the perisinusoidal space (of Disse).
SOM.MK.I.BPM2.3.DM.1.HCB.1290	Describe the microscopic structure of the perisinusoidal space (of Disse).
SOM.MK.I.BPM2.3.DM.1.HCB.1291	Describe the histological changes seen in the liver due to cirrhosis.
SOM.MK.I.BPM2.3.DM.1.HCB.1292	Describe the changes seen in fatty liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1293	Identify the gall bladder.
SOM.MK.I.BPM2.3.DM.1.HCB.1294	Give the function of the gall bladder.
SOM.MK.I.BPM2.3.DM.1.HCB.1295	Give the localization of the gall bladder.
SOM.MK.I.BPM2.3.DM.1.HCB.1296	Describe the microscopic of the gall bladder.
SOM.MK.I.BPM2.3.DM.1.HCB.1297	Describe the effects of cholecystokinin on the gall bladder.
SOM.MK.I.BPM2.3.DM.1.HCB.1298	Give the function of the intrahepatic billiary tree.
SOM.MK.I.BPM2.3.DM.1.HCB.1299	Describe the microscopic of the intrahepatic billiary tree.
SOM.MK.I.BPM2.3.DM.1.HCB.1300	Discuss liver necrosis in heart failure.
SOM.MK.I.BPM2.3.DM.1.HCB.1301	Give the localization of the pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1302	Give the function(s) of the pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1303	Compare exocrine vs. endocrine pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1304	Describe the histological appearance of the exocrine pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1305	Describe the microscopic structure of the pancreatic ducts.
SOM.MK.I.BPM2.3.DM.1.HCB.1306	Discuss the enzymes produced by the exocrine pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1307	Describe the effect of cholecystokinin on the exocrine pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1308	Describe the effect of secretin on the exocrine pancreas.
SOM.MK.I.BPM2.3.DM.1.HCB.1309	Describe the histological features observed in acute pancreatitis.
SOM.MK.I.BPM2.3.DM.1.HCB.1310	Describe the outcome in exocrine glands due to mutation in the gene that codes for CFTR protein.
DLA HCB	**Salivary glands**
SOM.MK.I.BPM2.3.DM.1.HCB.1190	Diffrentiate between the microscopic structures of the three major salivary glands.
SOM.MK.I.BPM2.3.DM.1.HCB.1191	Identify the microscopic structure of the ducts of the submandibular gland.
SOM.MK.I.BPM2.3.DM.1.HCB.1192	Identify the microscopic structure of the ducts of the parotid gland.
SOM.MK.I.BPM2.3.DM.1.HCB.1193	Give the function(s) of saliva.
SOM.MK.I.BPM2.3.DM.1.HCB.1194	Describe the composition of saliva.
Lecture BCHM	**Adipose Tissue Lipolysis, Fatty Acid Oxidation**
SOM. MK.I.BPM2.3.DM.2.BCHM.1193	Describe adipose tissue lipolysis and the role of hormone sensitive lipase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1194	Describe fatty acid activation
SOM. MK.I.BPM2.3.DM.2.BCHM.1195	Describe the carnitine shuttle mechanism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1196	Explain the consequences of defects in transport of fatty acids (carnitine deficiency and CPT-I and CPT-II deficiency)

SOM. MK.I.BPM2.3.DM.2.BCHM.1197	List the reactions of the mitochondrial β-oxidation pathway (not substrate names)
SOM. MK.I.BPM2.3.DM.2.BCHM.1198	Outline the energetics of β-oxidation
SOM. MK.I.BPM2.3.DM.2.BCHM.1199	Indicate the end products of oxidation of odd chain fatty acids
SOM. MK.I.BPM2.3.DM.2.BCHM.1200	Outline alpha-oxidation of branched chain fatty acids and indicate the metabolic defect in Refsum disease.
SOM. MK.I.BPM2.3.DM.2.BCHM.1201	Review the pathogenetic mechanism and metabolic changes in Jamaican vomiting sickness
SOM. MK.I.BPM2.3.DM.2.BCHM.1202	Compare and contrast systemic beta-oxidation defects and myopathic beta oxidation defects. Identify causes for each.
Lecture BCHM	**Ketone Body Metabolism and Disorders**
SOM. MK.I.BPM2.3.DM.2.BCHM.1203	Outline peroxisomal oxidation and omega-oxidation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1204	Discuss Zellweger syndrome as a disorder of peroxisomes
SOM. MK.I.BPM2.3.DM.2.BCHM.1205	Describe hepatic ketogenesis highlighting significance of mitochondrial HMG CoA synthase. List the major ketone body produced during hepatic ketogenesis .
SOM. MK.I.BPM2.3.DM.2.BCHM.1206	Explain ketone body utilization in peripheral tissues and significance of thiophorase
SOM. MK.I.BPM2.3.DM.2.BCHM.1207	Prepare a concept map indicating the steps involved in the generation of ketosis in starvation and uncontrolled type 1 diabetes mellitus.
SOM. MK.I.BPM2.3.DM.2.BCHM.1208	Correlate laboratory data in ketoacidosis (laboratory data in blood and urine) to the clinical signs in a type 1 diabetic
SOM. MK.I.BPM2.3.DM.2.BCHM.1209	Explain the biochemical basis of occurrence of hypoglycemia and hypoketosis in the systemic fatty acid oxidation disorders
Lecture PHYS	**Functions and Regulation of the GI Tract**
SOM.MK.II.BPM2.3.DM.1.PHYS.1095	Differentiate the processes of ingestion, digestion, absorption, secretion and excretion;include the location in the GI tract where each process occurs
SOM.MK.II.BPM2.3.DM.1.PHYS.1096	Outline the major anatomical organization of the enteric nervous system and the key cell types in enteric ganglia (sensory nerves, interneurons, and motor neurons).
SOM.MK.II.BPM2.3.DM.1.PHYS.1143	Discuss the role of the myenteric and submucosal plexi in regulation of gastrointestinal functions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1097	Explain how afferent and efferent extrinsic nerves (sympathetic and parasympathetic) interact with the enteric nervous system to regulate functions of the GI tract.
SOM.MK.II.BPM2.3.DM.1.PHYS.1098	Compare and contrast the regulation of gut function by nerves, hormones, and paracrine regulators.
SOM.MK.II.BPM2.3.DM.1.PHYS.1099	List the major excitatory and inhibitory motor neurotransmitters and digestive hormones in the GI tract and their key regulatory functions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1100	Analyze how GI cells integrate regulatory inputs and explain how the ultimate behaviour of GI tissues results from summation of inputs from multiple regulatory pathways. Explain how diet and nutrition influence appetite.
Lecture BCHM	**Cholesterol Metabolism and Regulation**
SOM. MK.I.BPM2.3.DM.2.BCHM.1210	Outline cholesterol synthesis and the role of acetylCoA as a precursor.
SOM. MK.I.BPM2.3.DM.2.BCHM.1211	Indicate the biochemical defect in Smith-Lemli-Opitz syndrome (SLOS).
SOM. MK.I.BPM2.3.DM.2.BCHM.1212	Identify the sources of NADPH for cholesterol synthesis: pentose phosphate pathway.
SOM. MK.I.BPM2.3.DM.2.BCHM.1213	Explain regulation of cholesterol synthesis at the level of HMG-CoA reductase.
SOM. MK.I.BPM2.3.DM.2.BCHM.1214	Outline bile acid synthesis and its regulation and bile salt formation.

SOM. MK.I.BPM2.3.DM.2.BCHM.1215	Describe the role of statins in the management of hypercholesterolemia
Lecture PHYS	**Enteric Motility**
SOM.MK.II.BPM2.3.DM.1.PHYS.1101	Discuss the esophageal pressure in terms of its normal resting range at different sites, method of measurements and variations with the respiratory cycle.
SOM.MK.II.BPM2.3.DM.1.PHYS.1102	Compare and contrast the structure and function of the upper versus lower esophageal sphincters.
SOM.MK.II.BPM2.3.DM.1.PHYS.1144	Explain the role of upper and lower esophageal sphincters in swallowing.
SOM.MK.II.BPM2.3.DM.1.PHYS.1103	Illustrate the dynamic pressure changes that occur in the esophagus after initiation of the swallowing reflex.
SOM.MK.II.BPM2.3.DM.1.PHYS.1104	Describe how dysfunction in the spatial or temporal characteristics of the esophageal pressure wave and/or sphincter relaxation can lead to swallowing defects and disorders such as heart burn, achalasia and aspiration of food.
SOM.MK.II.BPM2.3.DM.1.PHYS.1105	Describe the anatomical locations and role of interstitial cells of Cajal as slow wave pacemakers and mediators of inputs from the enteric nervous system.
SOM.MK.II.BPM2.3.DM.1.PHYS.1106	Explain the control of peristalsis by the enteric nervous system.
SOM.MK.II.BPM2.3.DM.1.PHYS.1107	Describe how extrinsic nerves (sympathetic and parasympathetic) affect GI motor patterns.
SOM.MK.II.BPM2.3.DM.1.PHYS.1108	Compare and contrast the functional roles of intrinsic and extrinsic enteric innervation in GI motor control.
SOM.MK.II.BPM2.3.DM.1.PHYS.1109	Describe major motor patterns in the GI tract and their functions during fasting (migrating motor complex or MMC) and during digestion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1110	List the afferent neuro-muscular pathways involved in the swallowing reflex, and the 2 major nuclei in the brain stem that integrate these motor pathways.
SOM.MK.II.BPM2.3.DM.1.PHYS.1111	Describe the storage, digestion, and motility roles of the stomach.
SOM.MK.II.BPM2.3.DM.1.PHYS.1112	Describe how distension of organs affects GI reflexes and alters responses to other regulatory inputs.
SOM.MK.II.BPM2.3.DM.1.PHYS.1113	Illustrate how luminal pressure and luminal stretch initiate motor reflexes in GI organs.
SOM.MK.II.BPM2.3.DM.1.PHYS.1114	Describe how these inputs (luminal pressure and luminal stretch) are integrated by intrinsic and extrinsic neural pathways including enteric ganglia, prevertebral ganglia, spinal cord and brain.
SOM.MK.II.BPM2.3.DM.1.PHYS.1115	Describe the function of colonic motility, in mediating formation of haustra and hasutral shuttling, mass movements through the transverse and distal colon, and defecation.
SOM.MK.II.BPM2.3.DM.1.PHYS.1116	Describe the sequence of events in the colon and anal sphincters occur ring during reflexive defecation, differentiating those movements under voluntary control and those under autonomic control.
Small group HCB	**DM SG 04 - Gastrointestinal Tract and Glands**
SOM.MK.I.BPM2.3.DM.1.HCB.1311	Identify and describe the cells, tissues, organs and associated structures of the gastrointestinal tract.
SOM.MK.I.BPM2.3.DM.1.HCB.1312	Present and demonstrate the cells, tissues, organs and associated structures of the gastrointestinal tract to the group.
SOM.MK.I.BPM2.3.DM.1.HCB.1313	Apply the histology of the gastrointestinal tract to clinical cases and correlations.
SOM.MK.I.BPM2.3.DM.1.HCB.1314	Identify and describe the cells, tissues, organs and associated structures of the gastrointestinal glands.
SOM.MK.I.BPM2.3.DM.1.HCB.1315	Present and demonstrate the cells, tissues, organs and associated structures of the gastrointestinal glands to the group.
SOM.MK.I.BPM2.3.DM.1.HCB.1316	Apply the histology of the gastrointestinal glands to clinical cases and correlations.
SOM.P.I.BPM2.5.DM.5.HCB.1317	To develop professional skills and presentation skills.

Lecture BCHM	**Ammonia Metabolism and Urea Cycle**
SOM. MK.I.BPM2.3.DM.2.BCHM.1222	Explain the formation of urea in the liver and its excretion via the kidneys.
SOM. MK.I.BPM2.3.DM.2.BCHM.1223	Compare and contrast the inherited disorders associated with the urea cycle highighting the enzyme deficient, laboratory findings, and clinical features of hyperammonemia.
SOM. MK.I.BPM2.3.DM.2.BCHM.1224	Identify the transport forms of ammonia from peripheral tissues (alanine & glutamine). Indicate the role of glutamate dehydrogenase and glutamine synthetase in glutamine formation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1225	Describe the transamination (aminotransferase) reaction and the role of vitamin B6 (pyridoxal phosphate).
SOM. MK.I.BPM2.3.DM.2.BCHM.1226	Describe the role of glutamate dehydrogenase in donating NH3 for the urea cycle. Identify the sources of nitrogen for the formation of urea
SOM. MK.I.BPM2.3.DM.2.BCHM.1227	List the reactions that form ammonia in the liver: Explain formation of ammonia in the intestine and its detoxification. Analyze the significance of intestinal ammonia formation in patients with compromised liver function.
SOM. MK.I.BPM2.3.DM.2.BCHM.1228	Indicate the metabolic basis for the use of phenylacetate, benzoic acid, arginine and carbamoyl glutamate (N-acetyl glutamate) in the management of the urea cycle disorders
SOM. MK.I.BPM2.3.DM.2.BCHM.1229	Analyze the mechanisms of neurotoxicity of hyperammonemia. Explain the metabolic basis of use of antibiotics and lactulose in the management of acquired hyperammonemia
DLA BCHM	**Introduction to Nitrogen Metabolism and Fates of Amino Acids**
SOM. MK.I.BPM2.3.DM.2.BCHM.1216	Analyze how amino acids may enter intermediary metabolism and describe 'amino acid pool'
SOM. MK.I.BPM2.3.DM.2.BCHM.1217	Discuss the reactions of alanine amino transferase, aspartate aminotransferase, glutaminase and glutamate dehydrogenase and analyze the central role of these amino acids in metabolism
SOM. MK.I.BPM2.3.DM.2.BCHM.1218	Define and enumerate the essential and nonessential amino acids in humans
SOM. MK.I.BPM2.3.DM.2.BCHM.1220	Distinguish the fates of C- skeletons of amino acids: Glucogenic amino acids, Ketogenic amino acids, Glucogenic & Ketogenic amino acids with examples.
SOM. MK.I.BPM2.3.DM.2.BCHM.1221	Review the TCA cycle intermediates formed from glutamate, glutamine, aspartate, and alanine.
Lecture PHYS	**Saliva and Gastric Secretions**
SOM.MK.II.BPM2.3.DM.1.PHYS.1117	Describe the volume and composition of salivary fluid coming from major salivary glands.
SOM.MK.II.BPM2.3.DM.1.PHYS.1118	Illustrate how acinar secretions are modified by ductal cells and predict the tonicity of the final secreted fluid depending on salivary flow rate.
SOM.MK.II.BPM2.3.DM.1.PHYS.1119	Describe the physiological function of the components of saliva.
SOM.MK.II.BPM2.3.DM.1.PHYS.1120	List the stimuli and neural pathways involved in promoting salivary secretion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1122	Analyse the 2 types gastric luminal fluid (oxyntic and non-oxyntic) and the mechanisms that mediate between food intake and oxyntic secretion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1123	Identify the proteins secreted into the gastric lumen by chief cells, parietal cells, And mucous cells. Contrast the functions and regulation of these secretions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1124	Discuss the gastric cell types secreting gastrin, somatostatin, histamine, and gastrin releasing peptide.
SOM.MK.II.BPM2.3.DM.1.PHYS.1145	Discuss the stimuli that promote and inhibit release of these peptides, and their cellular targets.

SOM.MK.II.BPM2.3.DM.1.PHYS.1125	Describe the luminal pH of the stomach in the basal fasted state versus the time course of changes in luminal pH after a mixed meal.
SOM.MK.II.BPM2.3.DM.1.PHYS.1126	Describe how parietal cells H+-K+-ATPase activity can be inhibited physiologically and pharmacologically.
SOM.MK.II.BPM2.3.DM.1.PHYS.1127	Describe the ion transport mechanisms and cellular enzymes needed to allow parietal cell homeostasis during gastric acid secretion.
Lecture BCHM	**Inherited Disorders of Amino Acid metabolism**
SOM. MK.I.BPM2.3.DM.2.BCHM.1230	Analyze deficiencies in biochemical pathways that are associated with inborn errors of metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1231	Discuss the appropriate use of language when referring to patients who have disorders with intellectual disability and delivery of findings to parents.
SOM. MK.I.BPM2.3.DM.2.BCHM.1232	Describe laboratory findings that are diagnostic for inborn errors of amino acid metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1233	Outline the catabolic pathway of phenylalanine and tyrosine and associated disorders like Phenylketonuria (Classic-PKU I and tetrahydrobiopterin deficiency – PKU II, maternal PKU), Alkaptonuria.
SOM. MK.I.BPM2.3.DM.2.BCHM.1234	Differentiate between the inborn errors of amino acid metabolism based on clinical features, laboratory findings, and vitamin/ coenzyme supplementation.
SOM. MK.I.BPM2.3.DM.2.BCHM.1235	Outline the catabolic pathway of branched chain amino acids and explain the metabolic basis for Maple syrup urine disease, Methylmalonic aciduria.
SOM. MK.I.BPM2.3.DM.2.BCHM.1219	Analyze and correlate clinical features and laboratory findings to the metabolic basis of cystinuria and Hartnup's disease
SOM. MK.I.BPM2.3.DM.2.BCHM.1236	Outline the catabolic pathway of methionine and cysteine and outline the two possible fates of homocysteine. Explain metabolic basis for homocystinuria.
Lecture PHYS	**Exocrine Pancreas and Intestinal Transports**
SOM.MK.II.BPM2.3.DM.1.PHYS.1128	List the major components secreted by the exocrine pancreas and the principal cell types involved in this secretion. Explain the role of pancreatic enzymes in digestion of dietary carbohydrates, proteins, and lipids.
SOM.MK.II.BPM2.3.DM.1.PHYS.1129	Describe the mechanisms by which chyme from the stomach is neutralized in the duodenum.
SOM.MK.II.BPM2.3.DM.1.PHYS.1130	List the stimuli that release secretin and CCK and explain the route by which these regulatory peptides stimulate the pancreas.
SOM.MK.II.BPM2.3.DM.1.PHYS.1131	Describe the role of CFTR in pancreatic ductal secretion, and predict the consequences of cystic fibrosis on the GI system.
SOM.MK.II.BPM2.3.DM.1.PHYS.1132	Describe the mechanisms by which HCO3- is secreted up by pancreatic ductal cells
SOM.MK.II.BPM2.3.DM.1.PHYS.1133	Describe the mechanisms whereby the gall bladder concentrates bile, and the endocrine mechanism stimulating gall bladder contraction and the secretion of bile through the sphincter of Oddi into the small intestine. Discuss role of bile in digestion and absorption of dietary lipids.
SOM.MK.II.BPM2.3.DM.1.PHYS.1134	Describe the location and the mechanisms that mediate the intestinal trans-epithelial movement of water, the major electrolytes, iron and calcium.
Small group BCHM	**DM SG 05 - Hypoglycemia and muscle weakness**
SOM. MK.I.BPM2.3.DM.2.BCHM.1378	Describe the steps of fatty acid β-oxidation, explain the role of the carnitine shuttle in mitochondrial fatty acid transport, and discuss the importance of carnitine in ATP generation from long-chain fatty acids.

SOM. MK.I.BPM2.3.DM.2.BCHM.1380	Describe the metabolic consequences of carnitine deficiency, including changes in energy metabolism and serum fuel profiles during fed and fasting states.
SOM. MK.I.BPM2.3.DM.2.BCHM.1384	Summarize the key steps, cellular locations, and regulation of fatty acid synthesis, β-oxidation, and ketogenesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1385	Describe the hormonal regulation of blood glucose homeostasis during feeding and fasting, focusing on the roles of insulin and glucagon.
SOM. MK.I.BPM2.3.DM.2.BCHM.1386	Describe the steps and regulation of glycogen synthesis and degradation, and compare glycogen metabolism between the liver and skeletal muscle.
SOM. MK.I.BPM2.3.DM.2.BCHM.1391	Describe the tissue-specific regulation of glycolysis in the liver and skeletal muscle, and integrate key aspects of carbohydrate metabolism including signal transduction, enzyme regulation, glycolysis, gluconeogenesis, glycogen metabolism, glycogen storage disorders, and blood glucose control.
Lecture BCHM	**Metabolic Roles of Folic Acid (B9) and Vitamin B12**
SOM. MK.I.BPM2.3.DM.2.BCHM.1255	Discuss the role of BH4 (tetrahydrobiopterin) in metabolism and effects of tetrahydrobiopterin deficiency (PKU II)
SOM. MK.I.BPM2.3.DM.2.BCHM.1256	Describe the clinical features and laboratory findings associated with folic acid deficiency, B12 deficiency, and inherited homocystinuria
SOM. MK.I.BPM2.3.DM.2.BCHM.1257	Explain mechanism of methyl-folate trap in B12 deficiency
SOM. MK.I.BPM2.3.DM.2.BCHM.1258	Specify one- carbon donors in metabolism and the groups they donate (SAM, THF, Cobalamin)
SOM. MK.I.BPM2.3.DM.2.BCHM.1259	Explain the formation and functions of S-Adenosyl methionine (SAM)
SOM. MK.I.BPM2.3.DM.2.BCHM.1260	Analyze metabolism of homocysteine, vitamins/ coenzymes required for metabolism & interpret clinical significance of homocysteine
SOM. MK.I.BPM2.3.DM.2.BCHM.1261	Explain formation of THF from folate and specify the mechanism of action of their inhibitors
SOM. MK.I.BPM2.3.DM.2.BCHM.1262	Summarize the role of folic acid as an acceptor and donor of 1-carbon groups in metabolism. Explain the role of folic acid as an acceptor of one carbon groups from amino acid metabolism and their utilization for nucleotide synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1263	Indicate different forms of THF (formyl, methylene and methyl) and reactions requiring different forms of THF
SOM. MK.I.BPM2.3.DM.2.BCHM.1264	Identify metabolic reactions requiring vitamin B12 as a coenzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1265	Compare and contrast causes, clinical and laboratory findings in folate and vitamin B12 deficiency
SOM. MK.I.BPM2.3.DM.2.BCHM.1266	Indicate role of intrinsic factor in vitamin B12 absorption
Lecture BCHM	**Digestion and Absorption**
SOM. MK.I.BPM2.3.DM.3.BCHM.1245	Outline digestion and absorption of dietary carbohydrates, proteins, lipids and nucleic acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1246	Outline digestion of dietary proteins and absorption of amino acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1247	Describe the proteolytic activation of zymogens using trypsinogen. Identify the specific cleavage sites of digestive proteases like trypsin, chymotrypsin and elastase.
SOM. MK.I.BPM2.3.DM.3.BCHM.1248	Describe absorption of monosaccharides into intestinal mucosal cells.
SOM. MK.I.BPM2.3.DM.3.BCHM.1249	Describe reaction catalyzed by pancreatic lipase and review importance of bile salts and colipase. Highlight their role in emulsification of dietary lipids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1250	Discuss the MAG pathway of TAG formation in mucosal cells. Outline formation and fate of chylomicrons.
SOM. MK.I.BPM2.3.DM.3.BCHM.1251	Describe enterohepatic circulation of bile salts. Summarize functions of bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.1252	Indicate mechanisms in various causes of steatorrhea (pancreatic disease, biliary tract obstruction, intestinal mucosal disease).

SOM. MK.I.BPM2.3.DM.3.BCHM.1253	Discuss bile composition and factors facilitating gallstone formation (cholelithiasis) and predict effect of bile duct obstruction on lipid digestion and absorption.
SOM. MK.I.BPM2.3.DM.3.BCHM.1254	Predict consequences of pancreatic disease (cystic fibrosis and pancreatitis) on digestion and discuss basis for dietary management.
DLA BCHM	**Digestion and Absorption**
SOM. MK.I.BPM2.3.DM.3.BCHM.1237	Describe digestion in mouth and stomach (role of salivary amylase and gastric lipase).
SOM. MK.I.BPM2.3.DM.3.BCHM.1238	Describe role of gastrin and activation of pepsinogen in stomach. Discuss consequences of insufficient acid production in stomach and the effect of antacids. Discuss gastric digestion of milk TAGs.
SOM. MK.I.BPM2.3.DM.3.BCHM.1239	Explain mechanisms involved in release of cholecystokinin and secretin. Review functions of cholecystokinin and secretin. Indicate role of HCO3- in pancreatic secretion and bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1240	Outline digestion and absorption of carbohydrates. Describe reactions of disaccharidases in brush border of intestinal cells.
SOM. MK.I.BPM2.3.DM.3.BCHM.1241	Describe lactose intolerance and differentiate primary, secondary and congenital lactose intolerance based on age of onset and primary defect.
SOM. MK.I.BPM2.3.DM.3.BCHM.1242	Outline lipid digestion and absorption and describe the significance of bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1243	Discuss differences between primary and secondary bile acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1244	Outline role of liver in synthesis of primary bile acids from cholesterol. Review formation of bile salts and their release into bile.
Lecture BCHM	**Heme Synthesis and Porphyrias**
SOM. MK.I.BPM2.3.DM.2.BCHM.1267	Describe significance of heme synthesis. Outline concept of heme synthesis starting from glycine and succinyl CoA to the formation of heme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1268	Distinguish between regulation of heme synthesis in liver and in erythroid cells.
SOM. MK.I.BPM2.3.DM.2.BCHM.1269	Describe reaction catalyzed by ALA synthase and explain how pyridoxine deficiency affects heme synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1270	Describe reaction catalyzed by ALA dehydratase and discuss effects of lead poisoning on heme synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1271	Classify porphyrias based on clinical manifestations such as photosensitivity, abdominal pain and neuropsychiatric manifestations.
SOM. MK.I.BPM2.3.DM.2.BCHM.1272	Describe metabolic basis for clinical features occurring in acute intermittent porphyria (AIP). Identify the deficient enzyme and products accumulating in blood and urine.
SOM. MK.I.BPM2.3.DM.2.BCHM.1273	Discuss the molecular basis for use of hemin in management of AIP.
SOM. MK.I.BPM2.3.DM.2.BCHM.1274	Explain the clinical significance of avoiding phenobarbital in patients with acute intermittent porphyria (AIP)
SOM. MK.I.BPM2.3.DM.2.BCHM.1275	Describe the metabolic basis for clinical features and laboratory findings in congenital erythropoietic porphyria. Identify the deficient enzyme and products accumulating in blood and urine.
SOM. MK.I.BPM2.3.DM.2.BCHM.1276	Describe the metabolic basis for clinical features of porphyria cutanea tarda. Identify the deficient enzyme and products accumulating in blood and urine.
Lecture BCHM	**Heme Degradation and Jaundice**
SOM. MK.I.BPM2.3.DM.4.BCHM.1277	Outline steps in degradation of heme to bilirubin in macrophages. Review the role of albumin in transport of bilirubin.

SOM. MK.I.BPM2.3.DM.4.BCHM.1278	Explain the consequence of a defect in uptake and conjugation of bilirubin in liver.
SOM. MK.I.BPM2.3.DM.4.BCHM.1279	Describe the fate of conjugated bilirubin in intestine. Review clinical significance of stercobilin and urobilin.
SOM. MK.I.BPM2.3.DM.4.BCHM.1280	Describe the conversion of heme to stercobilin and urobilin. Identify various tissues/organs involved in heme catabolism.
SOM. MK.I.BPM2.3.DM.4.BCHM.1281	Compare and contrast conjugated and unconjugated bilirubin with reference to chemical composition, water solubility, tissue deposition, excretion in urine and common clinical conditions where they are elevated
SOM. MK.I.BPM2.3.DM.4.BCHM.1282	Distinguish conjugated (direct) and unconjugated hyperbilirubinemia (indirect) using the Van den Bergh test.
SOM. MK.I.BPM2.3.DM.4.BCHM.1283	Explain mechanisms involved in development of physiological jaundice of newborn
SOM. MK.I.BPM2.3.DM.4.BCHM.1284	Explain rationale of phototherapy and phenobarbital in therapy for premature babies with jaundice.
SOM. MK.I.BPM2.3.DM.4.BCHM.1285	Distinguish prehepatic, hepatic and post hepatic jaundice based on laboratory findings and clinical features.
SOM. MK.I.BPM2.3.DM.4.BCHM.1286	Identify two disorders for prehepatic (hemolytic), hepatic and obstructive jaundice.
SOM. MK.I.BPM2.3.DM.4.BCHM.1287	Differentiate between inherited disorders: Gilbert syndrome, Dubin Johnson syndrome and Crigler-Najjar syndromes I and II based on age of presentation, pathogenetic mechanisms and metabolic alterations
Lecture BCHM	**Alcohol and Xenobiotic Metabolism in Liver**
SOM. MK.I.BPM2.3.DM.2.BCHM.1288	Describe role of cytochrome P450 enzymes in drug metabolism and specify roles CYP 2E1 and CYP 3A4. Discuss physiological relevance of induction of CYP 2E1 by ethanol.
SOM. MK.I.BPM2.3.DM.2.BCHM.1289	Describe mechanism of acetaminophen toxicity and the molecular basis for use of N-acetyl cysteine for management.
SOM. MK.I.BPM2.3.DM.2.BCHM.1290	Describe general features of ethanol and describe routes of hepatic ethanol metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1291	Discuss role of alcohol dehydrogenase, acetaldehyde dehydrogenases and microsomal ethanol oxidizing system (MEOS) in hepatic ethanol metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1292	Explain effects of high ethanol consumption on liver gluconeogenesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1293	Describe metabolic basis for acute and chronic toxic effects of ethanol abuse highlighting the role of acetaldehyde.
SOM. MK.I.BPM2.3.DM.2.BCHM.1294	Review the significance of drugs that inhibit aldehyde dehydrogenase (disulfiram) and alcohol dehydrogenase (fomepizole)
SOM. MK.I.BPM2.3.DM.2.BCHM.1295	Explain metabolic basis of fatty liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1296	Differentiate between methanol or ethylene glycol poisoning using laboratory findings (toxic compounds and anion gap) and clinical features. Explain the metabolic basis for management of these disorders.
Lecture BCHM	**Liver Function Tests**
SOM. MK.II.BPM2.3.DM.2.BCHM.1297	List important functions of the liver: Excretion of bilirubin, Synthesis of plasma proteins, Detoxification of ammonia
SOM. MK.II.BPM2.3.DM.2.BCHM.1298	Interpret the following laboratory tests in patients with acute hepatitis, alcoholic liver disease and cholestasis: Serum (total, conjugated and unconjugated) and urine bilirubin, Serum enzymes (ALT, AST, ALP, GGT), Serum proteins (albumin and globulin), Prothrombin time (INR), Serum ammonia, alpha fetoprotein (AFP), alpha1- antitrypsin, ceruloplasmin, serum iron, transferrin and ferritin
SOM. MK.II.BPM2.3.DM.2.BCHM.1299	Compare and contrast changes in liver function tests in hepatocellular diseases and disorders associated with cholestasis
SOM. MK.II.BPM2.3.DM.2.BCHM.1300	Differentiate between acute and chronic liver disease based on liver function tests

SOM. MK.II.BPM2.3.DM.2.BCHM.1301	Explain the molecular and pathophysiologic mechanisms underlying edema, icterus (jaundice), ascites, hepatic encephalopathy, and bleeding tendency in liver disease.
Small group BCHM;PHYS;HCB;ANAT	**DM SG 06 - Abdominal pain**
SOM. MK.I.BPM2.3.DM.2.BCHM.1210	Describe the major steps of cholesterol synthesis, including the role of acetyl CoA as the initiating substrate.
SOM. MK.I.BPM2.3.DM.2.BCHM.1214	Describe the pathway of bile acid synthesis from cholesterol, its regulation, and the formation and differentiation of primary and secondary bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.1242	Explain the biochemical steps involved in lipid digestion and absorption, emphasizing the role and recycling (enterohepatic circulation) of bile salts in these processes.
SOM. MK.I.BPM2.3.DM.3.BCHM.1253	Discuss the composition of bile, factors contributing to gallstone formation (cholelithiasis), and predict the consequences of bile duct obstruction on lipid digestion and nutrient absorption.
SOM. MK.I.BPM2.3.DM.3.BCHM.1392	Describe the structure and function of the inner mitochondrial membrane in the electron transport chain, identify major electron carriers in each ETC complex, and trace the path of electrons from NADH and $FADH_2$ to oxygen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1393	Describe the biochemical pathway of heme biosynthesis, including the required coenzymes, and explain the regulation of heme production in different tissues.
SOM. MK.I.BPM2.3.DM.2.BCHM.1394	Differentiate between acute intermittent porphyria, porphyria cutanea tarda, and congenital erythropoietic porphyria based on clinical features and laboratory findings, explaining the biochemical basis of each.
SOM. MK.I.BPM2.3.DM.2.BCHM.1395	Discuss how lead poisoning, pyridoxine (vitamin B6) deficiency, iron deficiency, and isoniazid therapy disrupt heme biosynthesis, and explain the underlying metabolic mechanisms.
SOM.MK.II.BPM2.3.DM.1.PHYS.1412	Compare and contrast the regulation of the gastrointestinal functions by neural, hormonal, and paracrine regulators.
SOM.MK.II.BPM2.3.DM.1.PHYS.1413	Differentiate major types of gastrointestinal motility patterns with respect to their site of origin, mechanism, and purpose in the gut.
SOM.MK.II.BPM2.3.DM.1.PHYS.1414	Examine the role of intrinsic (enteric nervous sytem) and extrinsic (sympathetic and parasympathetic nervous sytem) neuronal regulations of the gastrointestinal motility.
SOM.MK.II.BPM2.3.DM.1.PHYS.1415	Distinguish origin, integration, regulation, and purpose of various gastrointestinal reflexes.
SOM.MK.I.BPM2.3.DM.1.HCB.1318	Discuss the flow of bile from the hepatocytes where it is made to its exit in the duodenum
SOM.MK.I.BPM2.3.DM.1.HCB.1319	Microscopically describe the cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1320	Identify the parts of the biliary tree lined by cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1321	Discuss the function(s) of cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1322	Discuss the hormonal impact on the function of cholangiocytes
SOM.PB.I.BPM2.5.DM.5.HCB.1323	Generating a deeper understanding of professional and presentation skills
SOM.MK.1.BPM2.3.DM.1.ANAT.1167	Describe the location of the gall bladder and its relationship to the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1170	List the 3 common sites where gall stones can be impacted (lodged) and explain the effects of each on the bile flow.
SOM.MK.1.BPM2.3.DM.1.ANAT.1171	Explain how the visceral pain from an inflamed gallbladder can refer to the shoulder.
Lecture BCHM	**Water Soluble Vitamins**
SOM. MK.III.BPM2.3.DM.3.BCHM.1313	Compare and contrast fat-soluble and water-soluble vitamins regarding storage, absorption and risk factors for deficiency
SOM. MK.III.BPM2.3.DM.3.BCHM.1314	Discuss the metabolic basis, clinical manifestations and laboratory tests for thiamine deficiency (B1) in beri-beri and Wernicke Korsakoff syndrome.

SOM. MK.III.BPM2.3.DM.3.BCHM.1315	Indicate coenzyme forms, functions and clinical manifestations of riboflavin deficiency
SOM. MK.III.BPM2.3.DM.3.BCHM.1316	Recapitulate risk factors and metabolic basis of clinical manifestations in niacin deficiency (pellagra).
SOM. MK.III.BPM2.3.DM.3.BCHM.1317	Correlate metabolic basis for clinical manifestations of pyridoxine deficiency (B6). Review significance of B6 supplementation in patients on INH (isoniazid) therapy.
SOM. MK.III.BPM2.3.DM.3.BCHM.1318	Correlate metabolic basis for clinical features of vitamin C deficiency.
SOM. MK.III.BPM2.3.DM.3.BCHM.1319	Describe the utility of vitamin supplementation in management of patients with inherited metabolic disorders.
DLA BCHM	**Fat soluble vitamins (A, D, E and K)**
SOM. MK.III.BPM2.3.DM.3.BCHM.1302	Predict the reason for occurrence of fat-soluble vitamin deficiency in patients with fat malabsorption (cystic fibrosis, obstructive jaundice)
SOM. MK.III.BPM2.3.DM.3.BCHM.1303	Review the role of vitamin K in the maturation of clotting factors in the liver.
SOM. MK.III.BPM2.3.DM.3.BCHM.1304	Predict the risk factors (newborn, antibiotics), causes and effects of vitamin K deficiency.
SOM. MK.III.BPM2.3.DM.3.BCHM.1305	Discuss mechanism of inhibition of gamma-carboxylation by warfarin.
SOM. MK.III.BPM2.3.DM.3.BCHM.1306	Describe the functions and effects of dietary vitamin A deficiency. Discuss use of retinoids as drugs in medicine
SOM. MK.III.BPM2.3.DM.3.BCHM.1307	Explain why retinoic acid cannot be used to treat night blindness.
SOM. MK.III.BPM2.3.DM.3.BCHM.1308	Describe formation of vitamin D in skin and the formation of calcitriol (role of liver and kidney)
SOM. MK.III.BPM2.3.DM.3.BCHM.1309	Describe functions of 1,25 dihydroxy-D (calcitriol) related to calcium and phosphate metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1310	Indicate vitamins that act as steroid hormones (calcitriol and retinol)
SOM. MK.III.BPM2.3.DM.3.BCHM.1311	Indicate clinical features and manifestations of hypervitaminosis A and hypervitaminosis D.
SOM. MK.III.BPM2.3.DM.3.BCHM.1312	Describe risk factors and molecular basis of clinical features in rickets and osteomalacia.
Lecture BCHM	**The feed-Fast Cycle: Role of Hormones**
SOM. MK.III.BPM2.3.DM.3.BCHM.1320	Describe roles of insulin and glucagon and insulin:glucagon ratio in regulation of carbohydrate, lipid and protein metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1321	Describe roles of epinephrine and glucocorticoid in regulation of fuel metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1322	Differentiate between postprandial phase, post-absorptive phase and early phase of starvation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1323	Describe pathways leading to fatty synthesis and cholesterol synthesis in the fed state starting with glucose.
SOM. MK.III.BPM2.3.DM.3.BCHM.1324	Describe effect of changes in plasma epinephrine levels on carbohydrate, lipid and protein metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1325	Outline metabolic changes that occur during the feed-fast cycle.
SOM. MK.III.BPM2.3.DM.3.BCHM.1326	Describe changes in plasma glucose levels in the feed-fast cycle and during prolonged starvation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1327	Describe regulation and coordination of metabolic pathways during the feed-fast cycle in the various tissues.
DLA BCHM	**Obesity**
SOM. MK.III.BPM2.3.DM.3.BCHM.1328	Define obesity using body mass index (BMI)
SOM. MK.III.BPM2.3.DM.3.BCHM.1329	Compare the various methods available for diagnosis of obesity including BMI, waist circumference, Waist to hip ratio, bioelectric impedence and discuss the significance of location of body fat (‘apple’ vs ‘pear’) in obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1330	Summarize factors contributing (multifactorial) to obesity: including genetic, environmental, behavioral factors and the role of leptin.

SOM. MK.III.BPM2.3.DM.3.BCHM.1331	Analyze metabolic changes in obesity: including Dyslipidemia (Correlate changes in lipid profile to metabolic changes in obesity) and Syndrome X or Metabolic syndrome (Correlate clinical and biochemical features and explain significance of insulin resistance).
SOM. MK.III.BPM2.3.DM.3.BCHM.1332	Explain nutritional and lifestyle changes advised for the management of obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1333	Evaluate the role of surgery and drugs (sibutramine and orlistat) in obesity management.
Lecture BCHM	**Lysosomal Storage Disorders**
SOM. MK.I.BPM2.3.DM.3.BCHM.1334	Describe metabolic basis of hepatosplenomegaly, intellectual disability, macular cherry red spot in patients with lysosomal storage disorders.
SOM. MK.I.BPM2.3.DM.3.BCHM.1335	Distinguish between lysosomal storage diseases based on deficient enzyme, accumulating compound, clinical features, general cell appearance and clinical features: Hunter vs Hurler syndrome; Tay-Sachs disease; Fabry disease; Gaucher disease; Niemann-Pick disease; Metachromatic leukodystrophy; I-cell disease
Lecture BCHM	**Minerals and Nutritional Anemias**
SOM. MK.III.BPM2.3.DM.3.BCHM.1336	Indicate metabolic roles of trace minerals like iodine, zinc and selenium.
SOM. MK.III.BPM2.3.DM.3.BCHM.1337	Identify role of copper as a cofactor for enzymes (lysyl oxidase)
SOM. MK.III.BPM2.3.DM.3.BCHM.1338	Discuss clinical manifestations, laboratory findings and molecular defects in Wilson disease
SOM. MK.III.BPM2.3.DM.3.BCHM.1339	Review molecular basis of Menkes syndrome.
SOM. MK.III.BPM2.3.DM.3.BCHM.1340	Describe iron metabolism and regulation of its absorption
SOM. MK.III.BPM2.3.DM.3.BCHM.1341	Interpret significance of estimation of serum ferritin, serum iron, transferrin saturation and TIBC in iron deficiency anemia and hereditary hemochromatosis
SOM. MK.III.BPM2.3.DM.3.BCHM.1342	Discuss laboratory findings, inheritance and molecular basis of hereditary Hemochromatosis.
SOM. MK.III.BPM2.3.DM.3.BCHM.1343	Differentiate between nutritional causes of microcytic anemia (Iron, pyridoxine, copper deficiency and lead poisoning) using laboratory findings and clinical presentation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1344	Differentiate between nutritional causes of macrocytic anemia (vitamin B12 and folate) using clinical features and laboratory findings
Small group BCHM;PHYS	**DM SG 07 - Elevated Blood Ammonia**
SOM. MK.I.BPM2.3.DM.2.BCHM.1396	Explain the biochemical mechanisms leading to steatorrhea in pancreatic disease, biliary tract obstruction, and intestinal mucosal disorders.
SOM. MK.I.BPM2.3.DM.2.BCHM.1397	Predict the digestive consequences of pancreatic insufficiency (e.g., cystic fibrosis, chronic pancreatitis), and evaluate appropriate dietary and enzyme replacement strategies.
SOM. MK.I.BPM2.3.DM.3.BCHM.1301	Explain the molecular and pathophysiologic mechanisms underlying edema, icterus (jaundice), ascites, hepatic encephalopathy, and bleeding tendency in liver disease.
SOM. MK.I.BPM2.3.DM.3.BCHM.1299	Evaluate liver function tests by comparing patterns of abnormalities in hepatocellular versus cholestatic diseases, differentiating acute and chronic liver injury, and explaining their clinical diagnostic utility.
SOM. MK.I.BPM2.3.DM.3.BCHM.1398	Identify the causes of muscle weakness and wasting in chronic liver disease, including malnutrition, electrolyte disturbances, and metabolic dysfunction.
SOM. MK.I.BPM2.3.DM.3.BCHM.1399	Identify and localize the causes of malabsorption throughout the gastrointestinal tract, from the oral cavity to the colon.

SOM. MK.I.BPM2.3.DM.3.BCHM.1400	Define dehydration, explain its causes, and describe its physiological consequences.
SOM. MK.I.BPM2.3.DM.2.BCHM.1300	Differentiate between acute and chronic liver disease using patterns in liver function test results.
SOM. MK.I.BPM2.3.DM.2.BCHM.1402	Differentiate between hepatic and posthepatic causes of jaundice based on clinical features and laboratory findings.
SOM. MK.I.BPM2.3.DM.2.BCHM.1403	Explain the metabolic basis of drug-induced liver toxicity, highlighting toxic intermediates formed during metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1404	Describe the biochemical basis of the clinical manifestations observed in acute alcohol intoxication, and identify factors contributing to fatty liver and complications in chronic alcoholic cirrhosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1406	Describe the clinical features, laboratory findings and complications in chronic alcoholic cirrhosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1407	Explain the key biochemical reactions involved in nitrogen metabolism, including transamination and deamination reactions, their coenzymes, and their physiological significance.
SOM. MK.I.BPM2.3.DM.2.BCHM.1408	Describe the steps of the urea cycle, its metabolic regulation, and differentiate metabolic disorders associated with enzyme deficiencies at each step.
SOM. MK.I.BPM2.3.DM.2.BCHM.1409	Interpret laboratory findings to diagnose urea cycle disorders, create a differential diagnosis among urea cycle disorders and orotic aciduria, and analyze the metabolic fate of carbamoyl phosphate in ornithine transcarbamoylase deficiency.
SOM. MK.I.BPM2.3.DM.2.BCHM.1410	Create a differential diagnosis between all urea cycle disorders and orotic aciduria due to pyrimidine biosynthesis defect.
SOM. MK.I.BPM2.3.DM.2.BCHM.1412	Evaluate metabolic basis for treatment strategies for inborn urea cycle disorders.
SOM. MK.I.BPM2.3.DM.2.BCHM.1413	Evaluate the ethical considerations involved in genetic counseling for inborn errors of metabolism, including issues of reproductive decision-making, confidentiality, and informed consent.
SOM. MK.I.BPM2.3.DM.2.BCHM.1401	Evaluate the significance and utility of liver function tests.
SOM.MK.II.BPM2.3.DM.1.PHYS.1416	Evaluate major secretions by the exocrine pancreas with respect to their source, mechanism, composition, regulation, and functions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1417	Analyse functions of the gall bladder and discuss how they are integrated with the phases of digestion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1418	Compare and contrast the locations and the mechanisms that mediate the intestinal trans-epithelial movement of water, the major electrolytes, iron and calcium.
SOM.MK.II.BPM2.3.DM.1.PHYS.1419	Evaluate common causes of steatorrhea, and predict effects of steatorrhea on absorption of fat-soluble vitamins.
SOM.MK.II.BPM2.3.DM.1.PHYS.1420	Differentiate secretory and osmotic diarrhea and analyse the health consequences of diarrhea like fluid loss, electrolyte disturbances, and acid-base disturbances.
SOM.MK.I.BPM2.3.DM.1.HCB.1324	Discuss the microscopic changes observed in the liver as a result of fatty liver due to alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1325	Discuss the microscopic changes observed in the liver as a result of cirrhosis due to alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1326	State the cell type involved in the formation of fibrotic tissue as a result of alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1327	Discuss the normal microscopic structure of the cell type involved in the laying down of fibrotic tissue in the liver as a result of alcohol liver disease
SOM.MK.I.BPM2.3.DM.1.HCB.1328	Create a deeper appreciation for presentation skills and application of professionalism
Lecture BCHM	**Nutrition & Intensive Care**
SOM. MK.III.BPM2.3.DM.3.BCHM.1353	Generalize features of hypermetabolic response or Systemic inflammatory response syndrome (SIRS) and list its causes

SOM. MK.III.BPM2.3.DM.3.BCHM.1354	Explain the role of hormones (epinephrine, cortisol and insulin) and inflammatory mediators (cytokines and PGE2) in patients with critical illness.
SOM. MK.III.BPM2.3.DM.3.BCHM.1355	Correlate hormonal and metabolic changes (carbohydrate, protein and lipid) following trauma in the three phases: Ebb phase (Before resuscitation), Flow phase (Adrenergic-corticoid phase), Recovery phase (convalescent/ anabolic phase)
SOM. MK.III.BPM2.3.DM.3.BCHM.1356	Indicate significance of acute phase proteins in response to critical illness
SOM. MK.III.BPM2.3.DM.3.BCHM.1357	Explore significance of nutritional support and role of glutamine, arginine, antioxidants and omega-3 fatty acid supplementation (Immunomodulators and immunonutrition)
SOM. MK.III.BPM2.3.DM.3.BCHM.1358	Identify minerals (Copper and zinc) and vitamins (vitamin C) important for wound healing in post-operative patients
SOM. MK.III.BPM2.3.DM.3.BCHM.1359	Distinguish pros and cons of enteral vs parenteral nutrition
SOM. MK.III.BPM2.3.DM.3.BCHM.1360	Compare and contrast metabolic response to injury and starvation
Lecture BCHM	**Metabolic Changes in Diabetes Mellitus**
SOM. MK.I.BPM2.3.DM.3.BCHM.1361	Differentiate between type 1 and type 2 diabetes mellitus based on clinical presentation and laboratory findings.
SOM. MK.I.BPM2.3.DM.3.BCHM.1362	Correlate presenting features and metabolic basis for clinical features in patients with diabetes mellitus
SOM. MK.I.BPM2.3.DM.3.BCHM.1363	Explain metabolic basis for hyperglycemia while reviewing pathways and enzymes of carbohydrate metabolism that are active in liver in an uncontrolled diabetic.
SOM. MK.I.BPM2.3.DM.3.BCHM.1364	Explain metabolic changes in diabetic ketoacidosis (type 1 diabetics). Highlight occurrence of ketosis, metabolic acidosis (high anion gap) and hyperglycemia. Indicate changes in serum potassium following insulin therapy.
SOM. MK.I.BPM2.3.DM.3.BCHM.1365	Name the major ketone body found in blood and urine in diabetic ketoacidosis. Review enzymes of lipolysis, beta-oxidation and ketogenesis that are active in uncontrolled type 1 diabetic.
SOM. MK.I.BPM2.3.DM.3.BCHM.1366	Analyze metabolic basis for dehydration, hyperglycemia and hyperosmolarity in hyperosmolar hyperglycemic state in type 2 diabetics.
SOM. MK.I.BPM2.3.DM.3.BCHM.1367	Compare and contrast biochemical and laboratory findings in hyperosmolar hyperglycemic state and diabetic ketoacidosis
SOM. MK.I.BPM2.3.DM.3.BCHM.1368	Compare and contrast microvascular and macrovascular complications in diabetes mellitus
SOM. MK.I.BPM2.3.DM.3.BCHM.1369	Explain biochemical basis for mechanism of microvascular changes (sorbitol, AGE formation, role of DAG, hexosamine pathway) for nephropathy, neuropathy and retinopathy, Peripheral arterial disease and diabetic foot changes
SOM. MK.I.BPM2.3.DM.3.BCHM.1370	Identify changes in serum lipids (dyslipidemia) responsible for macrovascular complications
SOM. MK.I.BPM2.3.DM.3.BCHM.1371	Evaluate laboratory tests for management of diabetes mellitus. Analyze significance of microalbuminuria in identification of early nephropathy
Lecture PHYS; BCHM	**Flipped Class: Clinical Cases (Biochemistry and Physiology)**
SOM. MK.I.BPM2.3.DM.3.BCHM.1244	Outline role of liver in synthesis of primary bile acids from cholesterol. Review formation of bile salts and their release into bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1243	Discuss differences between primary and secondary bile acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1251	Describe enterohepatic circulation of bile salts. Summarize functions of bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.1249	Describe reaction catalyzed by pancreatic lipase and review importance of bile salts and colipase. Highlight its role in emulsification of dietary lipids.

SOM. MK.I.BPM2.3.DM.3.BCHM.1372	Describe metabolic basis for use of statins, ezetimibe and bile acid sequestering agents in hypercholesterolemia.
SOM.MK.II.BPM2.3.DM.1.PHYS.1135	Describe common causes of steatorrhea, and predict effects of steatorrhea on absorption of fat-soluble vitamins.
SOM.MK.II.BPM2.3.DM.1.PHYS.1136	Explain physiological basis for occurence of fat-soluble vitamin deficiency manifestations in patients with steatorrhea.
SOM.MK.II.BPM2.3.DM.1.PHYS.1137	Differentiate between the two major causes of diarhhea (secretory and osmotic), analyse the 3 key health consequences of diarrhea (fluid loss, acidosis, K+ loss) and relate these to the key components of oral rehydration therapy (ORT).
SOM. MK.I.BPM2.3.DM.3.BCHM.0252	Review the synthesis of primary bile salts from cholesterol. Describe the fate of conjugated primary bile salts in the intestine.
SOM. MK.I.BPM2.3.DM.3.BCHM.0251	Discuss differences between primary and secondary bile acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.0259	Describe enterohepatic circulation of bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.0257	Describe the role of pancreatic lipase, bile salts and colipase in lipid digestion and absorption. Highlight the role of bile salts in the emulsification of dietary lipids.
SOM. MK.I.BPM2.3.DM.3.BCHM.0393	Describe metabolic basis for use of statins, ezetimibe and bile acid sequestering agents in hypercholesterolemia.
Lecture BCHM	**Flipped Class: Clinical Correlates and Inborn Errors of Metabolism**
SOM. MK.I.BPM2.3.DM.4.BCHM.1373	Differentiate between inborn errors of metabolism that cause hypoglycemia
SOM. MK.I.BPM2.3.DM.4.BCHM.1374	Predict clinical features and laboratory tests used for inherited Urea cycle disorders
SOM. MK.I.BPM2.3.DM.4.BCHM.1375	Discuss various clinical situations which result in increased anion gap metabolic acidosis and lactic acidosis.
SOM. MK.I.BPM2.3.DM.4.BCHM.1376	Differentiate between various disorders of muscle energy metabolism
SOM. MK.I.BPM2.3.DM.4.BCHM.1377	Review acid-base disorders associated with gastric outlet obstruction (pyloric stenosis), diarrhea, diabetic ketoacidosis, lactic acidosis, ethylene glycol and methanol poisoning.